Safe Reinforcement Learning for Connected and Automated Vehicle Platooning

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***Abstract*—Connected and automated vehicle (CAV) platooning is emerging as a promising transportation technology, attributed to its potential for improving traffic efficiency and fuel economy. Meanwhile, this trend has led to increased demands and challenges for the decision-making of CAV platoons. In response, deep reinforcement learning (RL) methods have been recently adopted for CAV platooning, owing to their exceptional capacity to tackle control issues in complex and large-scale autonomous systems. However, the safety concerns associated with RL significantly impede its applicability in safety-critical systems. In light of this challenge, this study develops a novel safe RL framework for CAV platooning. The approach commences by incorporating the actor-critic RL framework to synthesize efficient control policies for CAV platooning. Specifically, a Lyapunov-based critic network is proposed in the actor-critic RL framework, ensuring the driving safety and stability of the CAV platoon. Finally, the effectiveness and computing efficiency of the proposed methodology are substantiated through a series of comprehensive simulation experiments.**

***Index Terms*—Connected and automated vehicles, vehicle platooning, safe reinforcement learning, Lyapunov stability**

I. Introduction

With the enhancement of vehicle connectivity and autonomy, vehicle platooning has aroused a great interest in the transportation systems [1]–[3]. Within the platooning paradigm, multiple CAVs are managed at a consistent velocity and in closely spaced formations. This approach can contribute to a significant reduction in road capacity utilization and air resistance, subsequently enhancing the driving safety, traffic efficiency, and fuel economy of transportation systems [4]–[7].

Given that controlling a CAV platoon presents more challenges compared to conventional vehicle dynamic control [8], [9], researchers have dedicated great efforts to developing advanced control methods tailored for CAV platooning. In [10]–[12], cooperative adaptive cruise control (CACC) techniques were formulated specifically for vehicle platooning. Nevertheless, robust control was widely utilized for CAV platooning as well. The authors in [13], [14] presented robust feedback controllers to manage connected vehicle platoons within uncertain traffic environments and employed the Lyapunov-Razumikhin approach to guarantee system string stability. Model predictive control (MPC) emerged as another topical strategy in the domain of CAV platooning. Within this scope, distributed MPC methods were developed to ensure the stability of heterogeneous vehicle platoons under varying communication topologies [15], [16]. These studies comprehensively investigated the impacts of network-related challenges such as time delay, packet dropout, and interruptions. Moreover, to deal with modelling uncertainties and system disturbances in vehicle platoons, stochastic MPC [17] and robust MPC [18] methodologies were also proposed respectively. Both of these strategies achieved excellent control results by balancing the trade-off between system performance and robustness to uncertainties. Finally, several other classical control methods, such as sliding mode control, optimal control, etc., were also extensively utilized for the CAV platooning [19], [20].

As outlined previously, a large number of classical control techniques have been developed for CAV platooning, resulting in great improvements in both the efficiency and security of CAV platoons. However, the control performances of classical methods heavily depend on the accuracy of the platoon system models. Describing these models is particularly challenging in complicated platooning scenarios characterized by a diverse range of vehicle types, varying road conditions, and intricate interactions between surrounding vehicles. In addition, as the number of vehicles in the platoon increases, there emerges a corresponding escalation in the complexity of control algorithms. This necessitates a greater allocation of computational resources and time to effectively implement the vehicle platoon control. Therefore, to tackle these challenges, the research community attempted to utilize machine learning strategies as an alternative approach to achieve more efficient CAV platooning.

Recently, a continuous stream of RL-based vehicle platooning strategies has emerged, owing to their capability to

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handle complex optimization tasks and adapt to varying environmental conditions [21]. In [22]–[24], a range of RL methods, including deep deterministic policy gradient (DDPG), deep Q-learning, etc., were developed for CAV platooning. Within these studies, comprehensive analyses were carried out through multiple simulations to examine the algorithm convergence, vehicle control performance, and platoon stability. Besides, RL-based platooning methods considering the traffic environment were fully investigated as well. Specifically, RL was adopted for cooperative control of vehicle platoons under mixed traffic environment in [25]. Additionally, RL-based methods had found extensive application in optimizing energy efficiency in CAV platoons as well. In [26], a novel communication proximal policy optimization (PPO) algorithm was proposed to reduce energy consumption with vehicle platoons. In the work presented in [27], an RL strategy designed with the DDPG framework was developed to regulate platoon energy consumption. Notably, this study introduced an innovative cloud platform to execute the control algorithm for CAV platoons, yielding significant energy-saving performance gains.

Thanks to the development of RL, the CAV platoon control performance and fuel efficiency have seen substantial improvements, even in scenarios involving heterogeneous vehicles and complex traffic conditions. However, the majority of research efforts in these works have primarily concentrated on enhancing RL control performance for vehicle platoons, often neglecting the aspects of system stability and driving safety. Specifically, CAV platoons represent typical safety-critical systems that demand rigorous safety and stability guarantees throughout the entire process of RL policy development and deployment.

Therefore, to more effectively achieve the control target, we develop a novel safe RL framework dedicated to CAV platooning. Initially, the actor-critic RL framework is adopted to train the control policy for vehicle platooning, incorporating some tricks derived from the DDPG methodology to bolster training performance. Then, a control Lyapunov function (CLF) is structured as the critic network to ensure system stability during the training and deployment processes. Finally, the control performance and computational speed of our proposed safe RL framework are compared with the MPC algorithm for CAV platooning.

The rest of this manuscript is organized as follows. Section II presents the problem statement of the safe RL for CAV platooning. Then, the detailed algorithm development are introduced in Section III. The comprehensive simulation results will be outlined in the Section IV. At last, the conclusion is demonstrated in Section V.

II. Problem Statement

This study primarily focuses on achieving CAV platoon stability and accurate longitudinal speed tracking for each vehicle. The platoon configuration typically comprises a leader vehicle ($i = 0$) and N follower vehicles ($i = 1, 2, \cdots, N$), all interconnected within a unidirectional communication topology. The brief scheme is shown in Fig 1. In order to provide a clear exposition of the CAV platooning problem and control targets, we will introduce the platoon dynamics and the related Markov decision process (MDP) model separately.

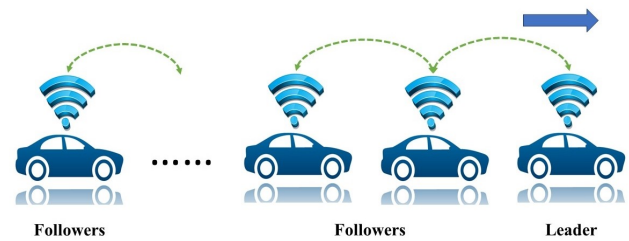

Fig. 1. Schematic diagram of the CAV platoon.

A. Platoon dynamics model

In the vehicle platoon, the position $p_i(t)$, velocity $v_i(t)$, and acceleration $a_i(t)$ of each vehicle are the most important states for platoon control. The platoon dynamics can be described as below:

$$\begin{cases} \dot{p}_i(t) = v_i(t) \\ \dot{v}_i(t) = a_i(t) \\ \dot{a}_i(t) = -\frac{1}{\tau_i} a_i(t) + \frac{1}{\tau_i} u_i(t) \end{cases}, \tag{1}$$

where τ_i denotes the drive-train dynamics coefficient depending on specific vehicle features, and $u_i(t)$ means the control input responsible for generating acceleration.

The main objective of the platoon control is to track the speed of the predecessor while maintaining a desired gap between any consecutive vehicles, which is specified by a desired spacing policy, that is:

$$\begin{cases} \lim\limits_{t\to\infty} \|e_{vi}(t)\| = \lim\limits_{t\to\infty} \|v_{i-1}(t) - v_i(t)\| = 0 \\ \lim\limits_{t\to\infty} \|e_{pi}(t)\| = \lim\limits_{t\to\infty} \|p_{i-1}(t) - p_i(t) - g\| = 0 \end{cases}, \tag{2}$$

where e_{vi} denotes velocity tracking errors between two adjacent vehicles, e_{pi} means the space tracking errors, and g is the desired constant space between adjacent vehicles.

To more effectively achieve the control target, we define $x_i(t) = [e_{pi}(t) \quad e_{vi}(t) \quad a_i(t)]^{\mathrm{T}}$. Then the system dynamic model is rewritten as:

$$\dot{x}_i(t) = A_i x_i(t) + B_i u_i(t) + D_i d_i(t), \tag{3}$$

$$A_i = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & -\frac{1}{\tau_i} \end{bmatrix}, B_i = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\tau_i} \end{bmatrix}, D_i = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \tag{4}$$

where $d_i(t) = a_{i-1}(t)$ is regared as the measured disturbance, which can be transmitted through the V2V network.

B. MDP model formulation

The vehicle platooning problem can be formulated as an MDP problem. The states, transition, actions, and reward functions are presented as follows.

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1) Transition of the state: The state space formula (3) can be discretized using forward Euler to model state transitions for RL. The observation state x_i for the MDP problem encompasses the ego-vehicle's position, velocity, acceleration, and the acceleration of the preceding vehicle.

2) Action space: Within a single-lane platoon, each vehicle should follow their predecessors with desired headways. The only actions for each vehicle are restricted to acceleration and deceleration. In the paradigm of RL, the action space is the same as the control inputs in (3).

3) Reward function: The design of a well-crafted reward function within the RL framework is of great importance to achieving high-performance outcomes in the system. Inappropriate reward design may lead to suboptimal or even undesirable behaviors. In the considered driving scenario, it is logical to minimize position error, velocity error and vehicle acceleration of the driving motion. Therefore, the proposed reward function incorporates three carefully selected elements: the normalized position error, e_{pi}, velocity error e_{vi} and vehicle acceleration a_i. Then the system reward $c(x, u)$ can be given below:

$$c(x,u) = -\sum_{i=1}^{N}(a\,\|e_{pi}\| + b\,\|e_{vi}\| + c\,\|a_i\|), \tag{5}$$

where $\|\cdot\|$ denotes the Euclidean norm, subscript x and u represent the vector of each vehicle state x_i and input u_i, a, b and c are the weight parameters.

III. Algorithm Development

To address the challenge of ensuring stability in the platoon system, a novel safe actor-critic RL algorithm to learn the control policies with guaranteed stability is proposed. We start by introducing the Lyapunov critic function V_c and provide a detailed explanation of its construction process. Subsequently, we integrate this Lyapunov critic function into the policy gradient formulation based on the actor-critic RL framework.

A. Lyapunov Critic Network Formulation

In this section, our emphasis is on the simultaneous learning of both the individual Lyapunov function and the vehicle platooning policy. Within the framework, the Lyapunov critic function V_c serves a dual purpose: ensuring stability and guiding policy learning. In order to empower actor-critic learning, the Lyapunov critic network is developed relying on the state x and the control input u, such that it can be exploited in judging the training performance of the actor. What's more, the critic network must adhere to the fundamental Lyapunov stability conditions, providing guidance to the actor network for exploration and exploitation within the stability region. The Lyapunov stability conditions are shown below:

$$-\alpha_1 c(x) \le V(x) \le -\alpha_2 c(x), \tag{6}$$

$$\nabla V(x) f(x) \le -\alpha_3 V(x), \tag{7}$$

where $c(x)$ denotes the step reward defined before, α_1, α_2, and α_3 are scalable positive constants, $\nabla V(x)$ means the Lie derivative of the Lyapunov function $V(x)$, and $f(x)$ stands for the system dynamics described in (3). Given that $c(x)$ is defined as a negative reward, a minus sign is used in this function to guarantee the positivity of both the left and right sides of the inequality.

To uphold the non-negativity constraint, the construction of the Lyapunov critic function V_c involves mapping state-action pairs within $V_c : X \times U \to \mathbb{R}_+$ to the non-negative real domain. A specific parameterization technique is employed within the Lyapunov critic network. The model details are depicted in the following exposition:

$$V_c(x,u) = f_\phi(x,u)^T f_\phi(x,u), \tag{8}$$

where f_ϕ represents a fully connected deep neural network with parameter ϕ, which acts as the heart of the parameterization and ensures the positive definiteness of $V_c(x, u)$.

Practically, we need to establish the conditions based on data collected during the interactive procedure. At each step, the agent would apply a control action and then receive feedback (reward or cost). To improve the training performance, a replay buffer $\mathcal{D}$ is first constructed to store the action-feedback pairs. After that, a batch of transition pairs is randomly sampled at each timestep to train the control Lyapunov function.

Then, according to the Lyapunov stability conditions described in (7) and the Lagrange relaxation technique, critic network V_c is updated using gradient descent by minimizing the following objective function:

$$\begin{aligned} J(\phi) =& \mathbb{E}_{\mathcal{D}}[(c(x,u) + \gamma V_c(x', \pi_\theta(x')) - V_c(x,u))^2 + \\ & \lambda(\nabla V_c(x,u) f(x,u) + \alpha_3 V_c(x,u) + \epsilon)], \end{aligned} \tag{9}$$

where $\mathbb{E}_{\mathcal{D}}[\cdot]$ expresses the expected value of the objective function using data sampled from the replay buffer $\mathcal{D}$, $\pi_\theta(\cdot)$ is the policy network and $\pi_\theta(x')$ means the calculated actions in the next step. λ denotes the Lagrange multiplier that controls the relative importance of the inequality condition, ϵ represents the small parameter that encourages strict satisfaction and generalization ability and γ indicates the discount factor.

B. Policy Learning Framework

This subsection delves into learning the control policy within the Lyapunov-based actor-critic RL framework. The policy learning problem is reformulated as an optimization task, and derived according to the following objective function:

$$J(\theta) = \mathbb{E}_{\mathcal{D}}(V_c(x,u)\big|_{u=\pi_\theta(x)}). \tag{10}$$

Then based on the objective function introduced above, we can update the actor network parameters through stochastic gradient descent. The approximated derivative of the objective function is shown as follows:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\mathcal{D}}(\nabla_u V_c(x,u)\big|_{u=\pi_\theta(x)} \nabla_\theta \pi_\theta(x)). \tag{11}$$

After obtaining the gradient of the objective function, the actor network parameters are updated as below:

$$\theta \leftarrow max(0, \theta + \sigma \nabla_\theta J(\theta)), \tag{12}$$

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where δ indicates the parameter learning rate.

Finally, the value of Lagrange multiplier λ also needs to be updated at each step to ensure the iteration loop runs effectively. The update is implemented by maximizing the following objective function using gradient descent.

$$J(\lambda) = \lambda \mathbb{E}_{\mathcal{D}}(\nabla V(x,u) f(x,u) + \alpha V(x,u) + \epsilon). \tag{13}$$

During the training process, the Lagrange multipliers are updated as follows:

$$\lambda \leftarrow max(0, \lambda + \delta \nabla_{\lambda} J(\lambda)), \tag{14}$$

where δ stands for the learning rate. The pseudo-code of the proposed Lyapunov-based safe RL algorithm is shown in Algorithm 1.

Algorithm 1 Lyapunov-based safe RL

Require: Maximum episode length K; maximum iteration steps M; Lyapunov critic network $V_{c,\phi}$; actor network π_θ; replay buffer $\mathcal{D}$; Lagrange multipliers λ

Repeat
  Receive the initial observation x_0
  for $i = 1$ to K **do**
    Sample u from $\pi_\theta(x|u)$ and step forward
    Observe x', c and store (x, u, c, x') in $\mathcal{D}$
  end for
  for $j = 1$ to M **do**
    Sample mini-batches of transitions from $\mathcal{D}$
    Update the Lyapunov critc network by optimizing Eq. (9)
    Update the actor network with Eq. (12)
    Update the Lagrange multipliers with Eq. (14)
  end for
  return $V_{c,\phi}$ and π_θ

IV. Experiment Results

In this section, we illustrate simulated CAV platoon control problem to demonstrate the general applicability of the proposed method. The effectiveness of the proposed framework will be analyzed according to the following aspects:

1) Training convergence: does the proposed training algorithm converge with random parameter initialization;

2) Control performance: can the goal of the task be achieved or the cumulative cost be maximized;

3) Online computing efficiency: can the trained control policy be executed online with high computing speed;

In this section, we evaluate the performance of the Lyapunov-based safe RL on the vehicle platooning task in comparison with MPC algorithms. In the first part, the training convergence performance of our developed framework is presented. The result potentially indicates our method is able to converge to the desired target. Then the control performance and computing efficiency of the designed safe RL will be compared with MPC, which is widely applied for vehicle platooning. The experiment results demonstrate the superiority of our learning control framework.

A. Training convergence

In this section, Fig. 2 showcases the average total award and its variance during the training procedure. As is seen in the figure, the average reward of our designed Lyapunov-based safe RL algorithm approaches the region near zero around 200 steps, which indicates the algorithm has a good convergence performance. In the training process, the reward of the Lyapunov-based RL exhibits some initial oscillations. This behavior is primarily attributed to the constraints imposed by Lyapunov stability, which may influence the optimal selection of control actions.

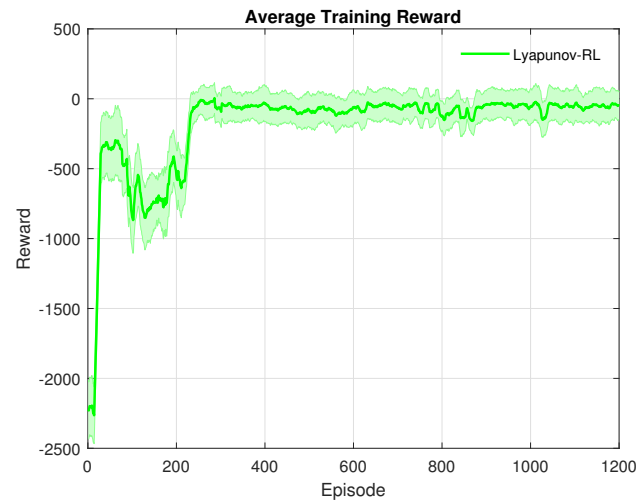

Fig. 2. Training performance of the Lyapunov-based safe RL.

B. Control performance

In this section, we mainly present the comparison of control performance between our proposed Lyapunov-based RL and MPC. The simulation cycle includes the vehicle starting stage, constant-speed driving stage, and varying-speed driving stage. The acceleration of the leader vehicle is pre-defined, and the follower vehicles are controlled through the proposed safe RL strategy. Besides, only the longitudinal dynamics are considered in this case, while the vehicle lateral dynamics are ignored.

The comparison of vehicle platoon accelerations between the Lyapunov-based RL and MPC is shown in Fig. 3. As is seen in this figure, our developed Lyapunov-based RL method has obviously better tracking performance at the initial stage, while the MPC strategy has bigger overshoots during the tracking process. In addition, at the varying-speed driving stage, the Lyapunov-based RL method could follow the acceleration of the leader vehicle accurately, but the MPC strategy does not demonstrate a very good tracking performance. Fig. 4 presents the comparison of vehicle platoon velocity between Lyapunov-based RL and MPC. This figure illustrates similar control outcomes to Fig. 3, reaffirming that our proposed Lyapunov-based RL method outperforms MPC in terms of vehicle platoon speed tracking.

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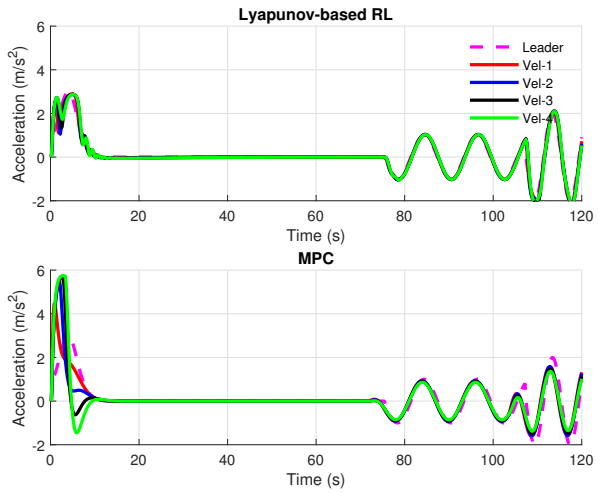

Fig. 3. Comparison of vehicle platoon accelerations between Lyapunov-based RL and MPC.

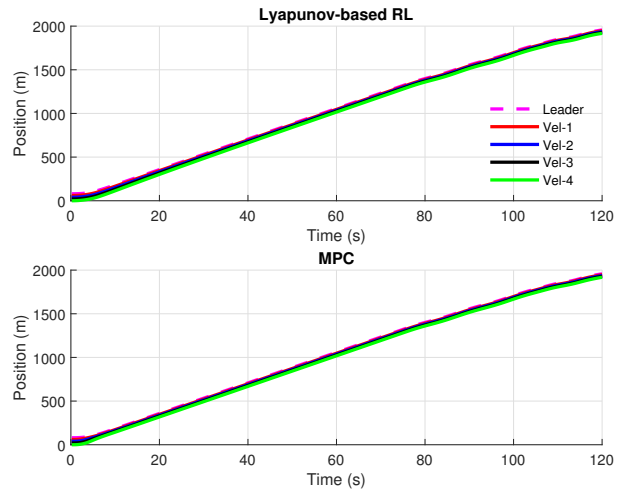

Fig. 5. Comparison of vehicle platoon driving positions between Lyapunov-based RL and MPC.

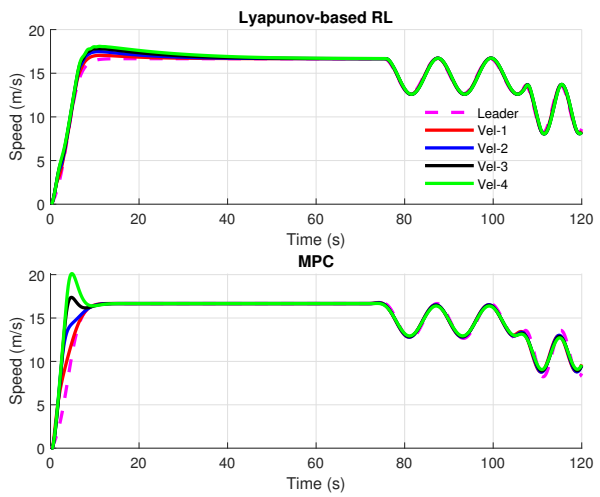

Fig. 4. Comparison of vehicle platoon velocity between Lyapunov-based RL and MPC.

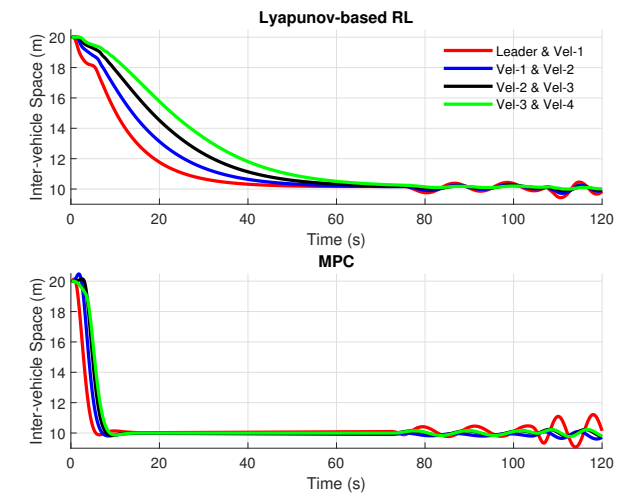

Fig. 6. Comparison of vehicle platoon inter-distance between Lyapunov-based RL and MPC.

Fig. 5 and Fig. 6 describe the comparison of platoon positions and vehicle inter-distance between Lyapunov-based RL and MPC, respectively. Specifically, the comparison of vehicle platoon inter-distance serves as a demonstrative measure of the control performance of our designed Lyapunov-based RL framework in contrast to the MPC strategy. As we can see in Fig. 6, both the Lyapunov-based RL and MPC methods effectively control the platoon inter-distance to converge to the desired setpoint. It's worth noting that MPC has a faster convergence speed compared with the Lyapunov-based RL, but it comes at the cost of control oscillations and driving discomfort. Nevertheless, the proposed Lyapunov-based RL also perform better control stability compared with MPC at the varying-speed driving stage. The Lyapunov-based RL method could keep the control error of each vehicle within 0.5m, while the MPC has a relatively large control error.

C. Online computing efficiency

The online computing efficiency is a significant advantage of RL for controlling nonlinear systems in comparison to MPC. In the RL framework, training the control policy networks can be time-consuming. However, once the policy networks are trained, deploying them for system control becomes straightforward and does not require any additional online computing. In contrast, MPC does not require pretraining of the control policy but relies on extensive real-time optimization, which can significantly increase the online computational burden.

To assess the online computing efficiency, several evaluation experiments are conducted. The average online computing time for both algorithms is recorded in Table I. These simulation experiments are performed using the Python platform on a computer equipped with an Intel(R) Core(TM) i7-10510U

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CPU. The values in Table I represent the actual computer execution time required for the 120s platoon control simulation.

As indicated in Table I, the online computing time of RL is considerably shorter than that of MPC. This observation demonstrates the significant potential of the RL method for real-time implementation in practical vehicle platooning scenarios.

TABLE I
THE COMPARISON OF ONLINE COMPUTING TIME FOR LYAPUNOV-BASED RL AND MPC.

Algorithm	Lyapunov-based RL	MPC
Computing time	4.2471s	127.6537s

V. CONCLUSION

This paper presents a novel approach to regulating the CAV platoons via a safe RL framework with Lyapunov stability from control theory. By incorporating the CLF as a critic network, an actor-critic RL framework is established to learn a control policy ensuring the vehicle platoon control performance and system stability simultaneously. Comprehensive testing across diverse scenarios reveals that our method not only achieves comparable convergence performance but also outperforms impressively traditional control approaches such as MPC. For future work, it might be interesting to extend this method to CAV platooning considering malicious cyber-attacks.

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